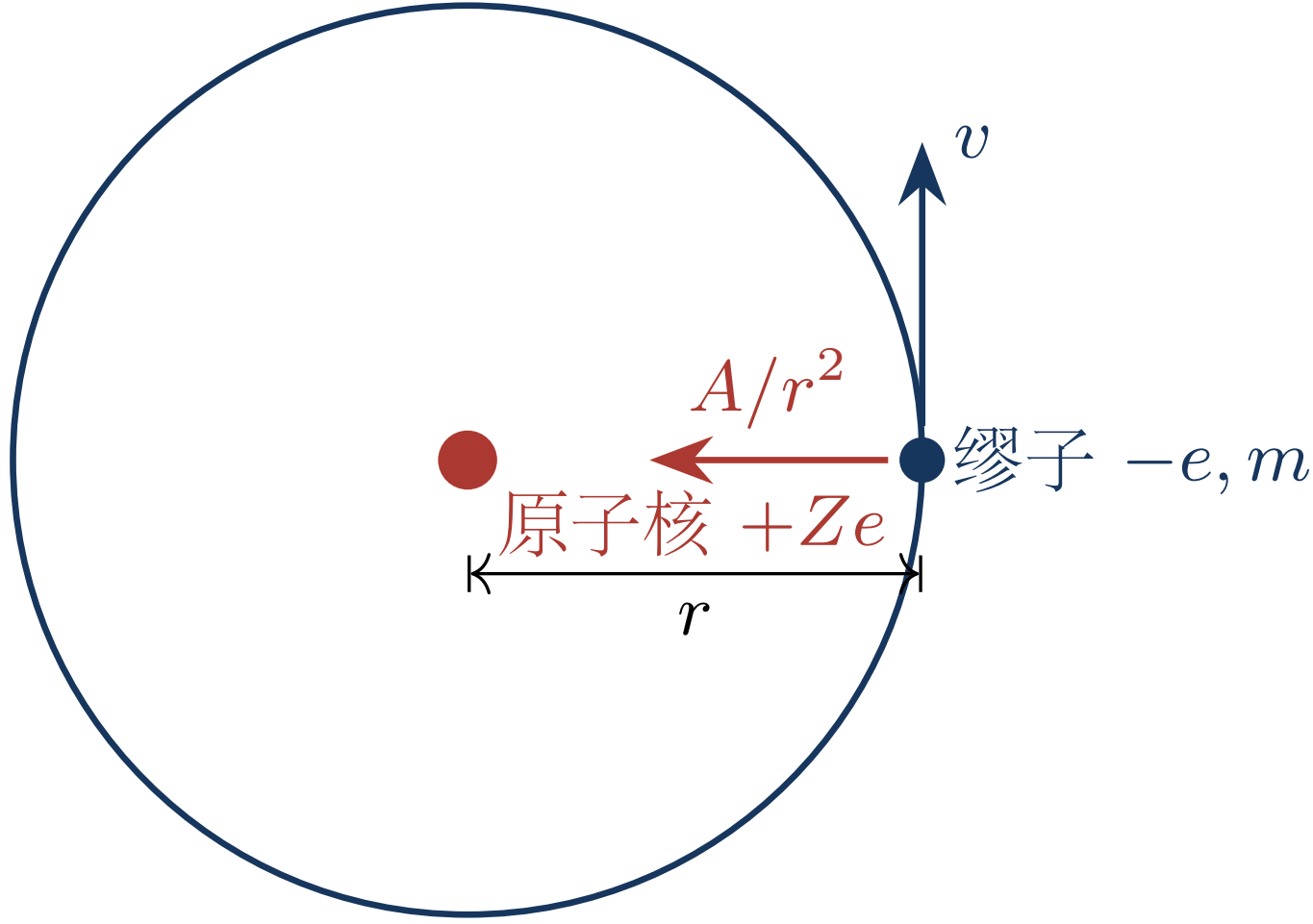
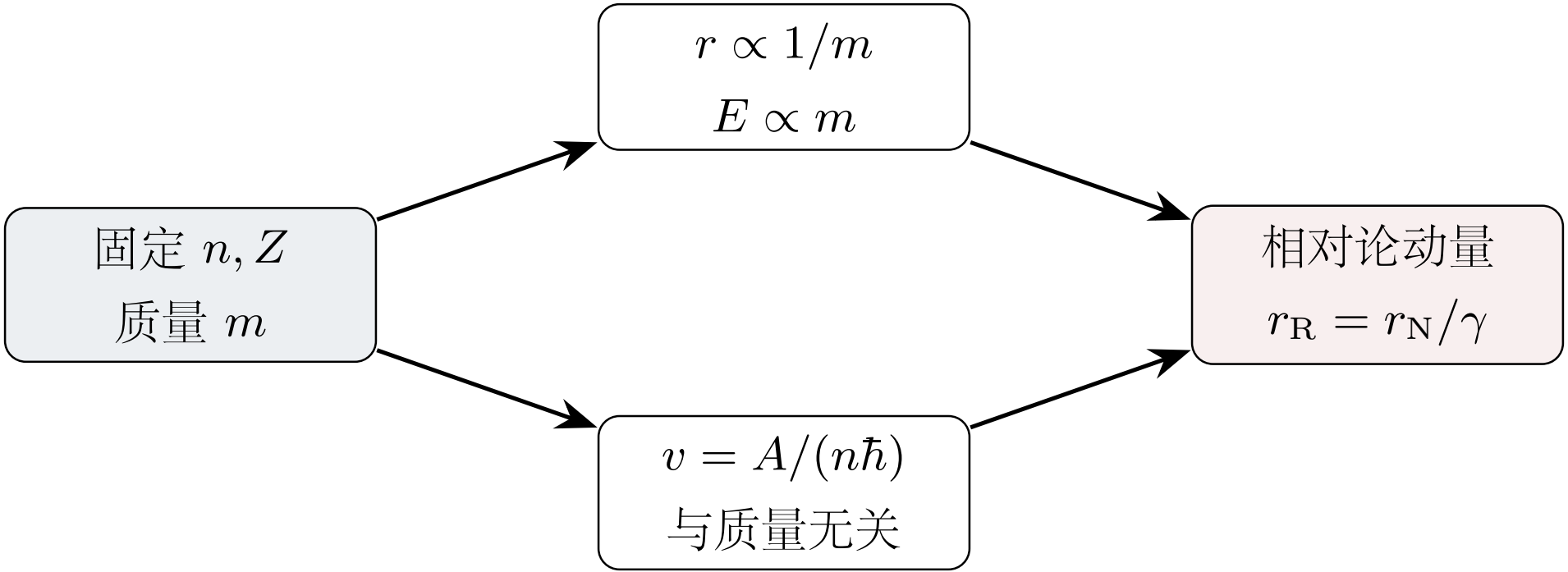
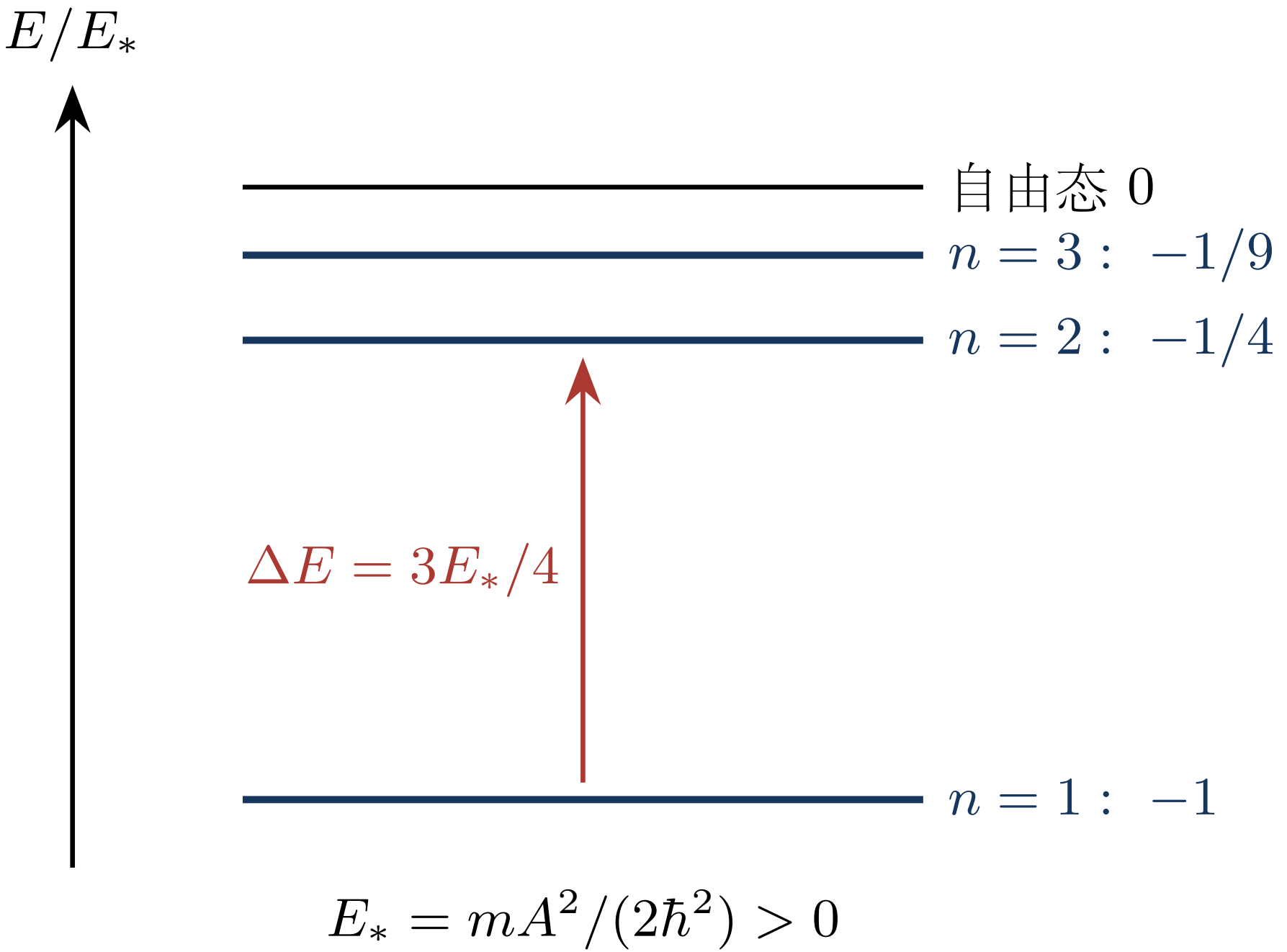
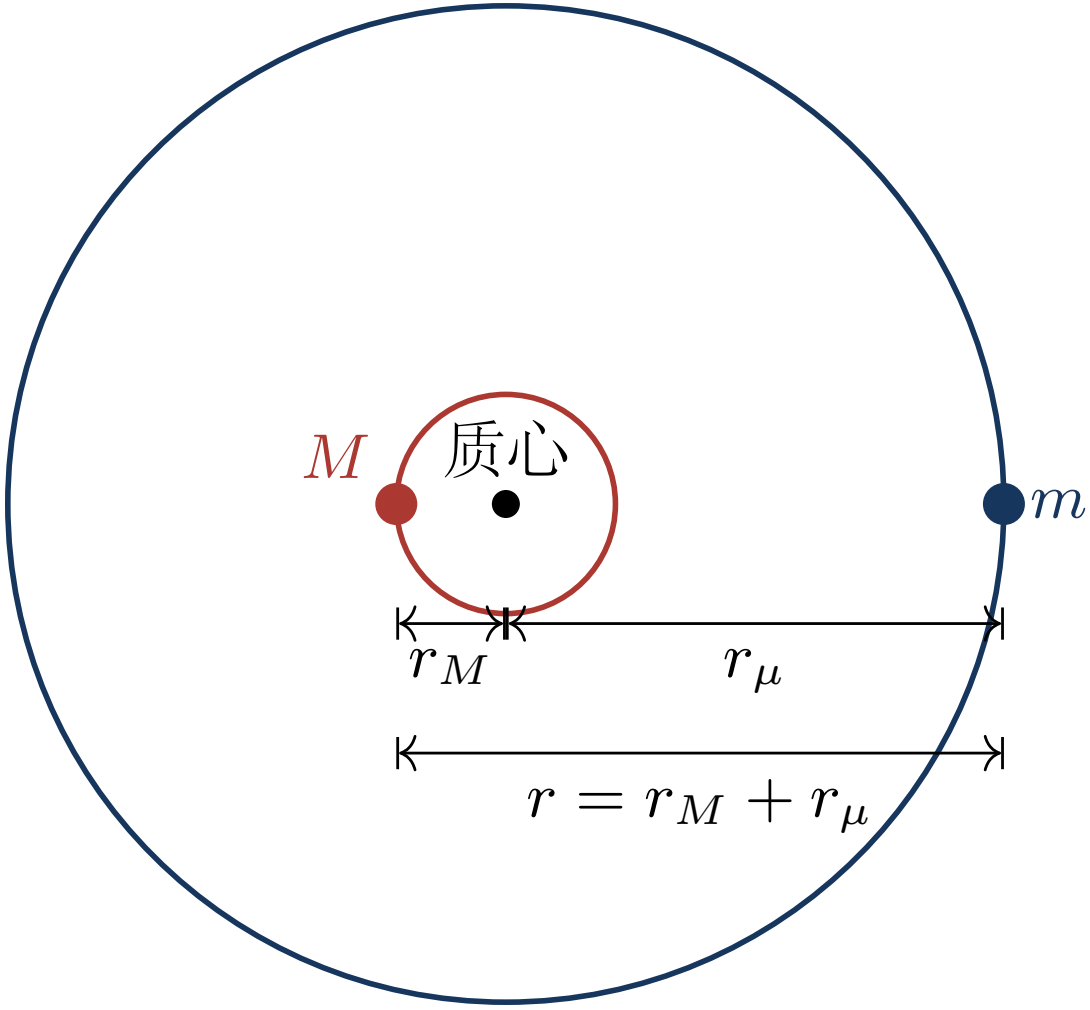
固定核近似

$$mvr = n\hbar$$



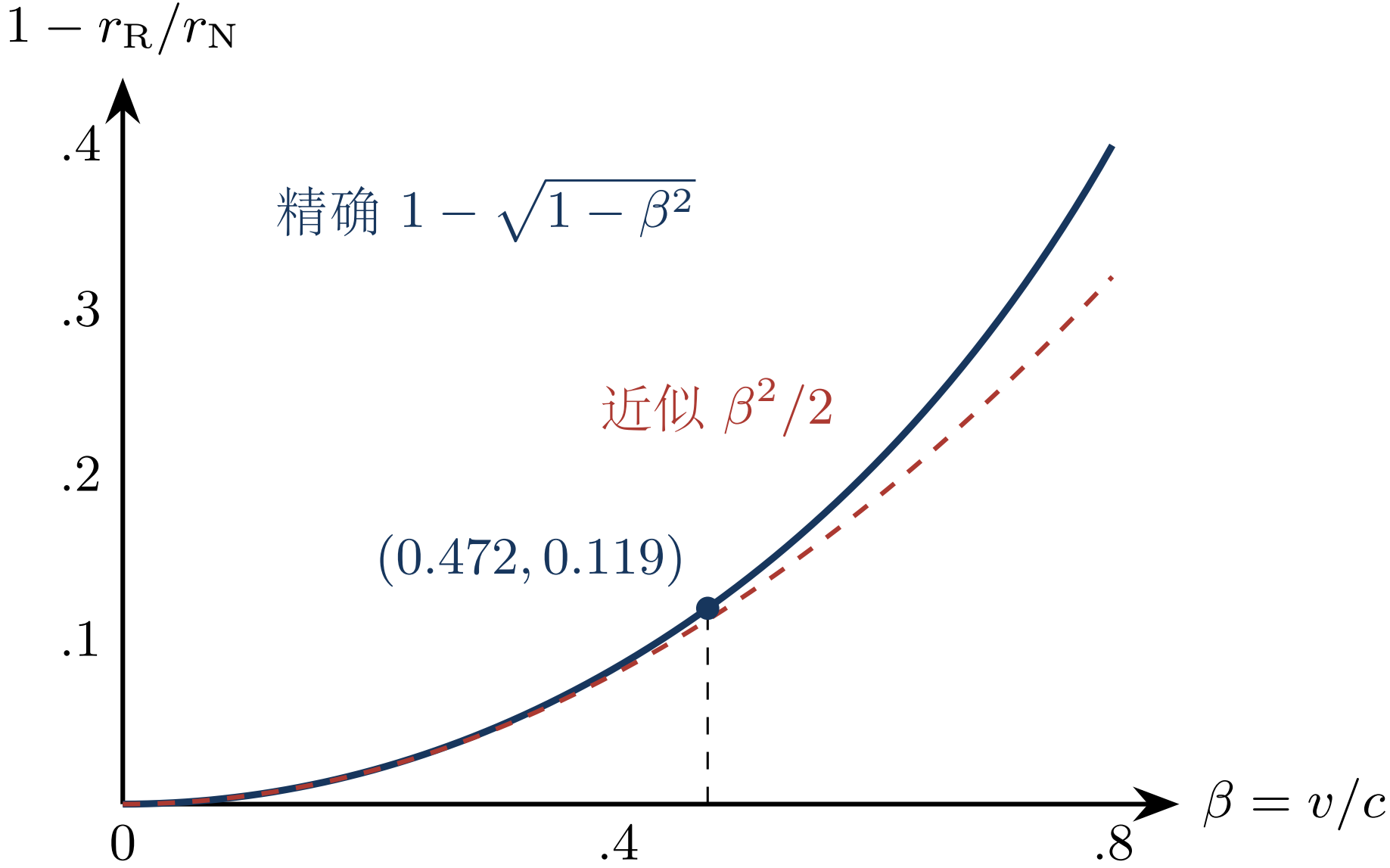

固定 n, Z
质量 m
$r \propto 1/m$
$E \propto m$
$v = A/(n\hbar)$
与质量无关
相对论动量
$r_{\mathrm{R}} = r_{\mathrm{N}}/\gamma$


E/E_*
自由态 0
$n=3:\ -1/9$
$n=2:\ -1/4$
$\Delta E = 3E_*/4$
$n=1:\ -1$
$E_* = mA^2/(2\hbar^2) > 0$



$$Mr_M = mr_\mu$$

$$\mu_r = mM/(m+M)$$


$1 - r_{\mathrm{R}}/r_{\mathrm{N}}$
.4
.3
.2
.1
精確 $1 - \sqrt{1 - \beta^2}$
近似 $\beta^2/2$
$(0.472, 0.119)$
0
.4
.8
$\beta = v/c$